

Discrete Mathematics Quiz 1

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Write clear and concise solutions. All problems carry equal marks.

1. Let $P(x_1, x_2, \dots, x_n)$ be a nonzero n -variate polynomial with coefficients that are positive integers. Let $\alpha_i, 1 \leq i \leq n$ be infinite cardinal numbers. Show that $P(\alpha_1, \alpha_2, \dots, \alpha_n)$ is an infinite cardinal number that equals α_i for some i .
2. Let $\alpha \leq \beta$ be two infinite cardinal numbers and A be a set of cardinality β . Show using Zorn's lemma that we can write A as a disjoint union of sets $B_i, i \in I$ for some index set I , such that each B_i has cardinality α .
3. Suppose $G = (V, E)$ is an infinite graph. Let $\bar{G} = (V, \bar{E})$ denote its complement. That is, $\bar{E}$ is $\binom{V}{2} \setminus E$. Show that either G or $\bar{G}$ has an infinite complete subgraph.
4. Let $f(x)$ denote the generating function for the sequence $\{f_n\}_{n \geq 0}$. Describe the sequence $\{g_n\}_{n \geq 0}$ generated by $\frac{f(x)}{(1-x)}$ in terms of the sequence $\{f_n\}_{n \geq 0}$. Using this show that $\sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6}$. **note: any other proof here will not get credit.**